Summary

Simon Gustafsson

12th of January 2025

1 Trigonometry

$$1 + \tan^2 x = \frac{1}{\cos^2 x} \iff \frac{1}{\cos^2 x} - 1 = \tan^2 x$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2} \implies \cos\frac{x}{2} = \pm\sqrt{\frac{1 + \cos x}{2}}$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2} \implies \sin\frac{x}{2} = \pm\sqrt{\frac{1 - \cos x}{2}}$$

$$\sin x \leqslant x \leqslant \tan x$$

$$\frac{x-1}{x} \leqslant \ln x \leqslant x - 1$$

2 Sequences

$$\sum_{k=0}^{n} x^k = \frac{1}{1-x} - \frac{x^{n+1}}{1-x}$$

Converges absolutely: $\left|a_n\right|$ converges $\implies a_n$ converges.
Converges conditionally: a_n converges, but not $\left|a_n\right|$.

2.1 Sum Convergence Tests

2.1.1 Divergence Test

If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

2.1.2 Comparison Test

If $\forall n \in \mathbb{N}, a_n > 0$ and $b_n > 0$, and $a_n > b_n$ for all $n \in \mathbb{N}$:

$$\sum_{k=1}^{\infty} a_k < \infty \Longrightarrow \sum_{k=1}^{\infty} b_k < \infty,$$

$$\sum_{k=1}^{\infty} b_k = \infty \Longrightarrow \sum_{k=1}^{\infty} a_k = \infty.$$

2.1.3 Alternating Series Test

If $\forall p \in \mathbb{N}$, $a_{2p} > 0$ and $a_{2p-1} < 0$ (or vice versa), and $|a_n|$ is decreasing, then $\sum_{k=1}^{\infty} a_k$ converges.

2.1.4 Limit Comparison Test

If

$$\lim_{n\to\infty} \frac{a_n}{b_n} \in (0, \infty),$$

then:

$$\sum_{k=1}^{\infty} a_k < \infty \Longleftrightarrow \sum_{k=1}^{\infty} b_k < \infty,$$

$$\sum_{k=1}^{\infty} a_k = \infty \Longleftrightarrow \sum_{k=1}^{\infty} b_k = \infty.$$

2.1.5 Root Test

Let $\lim_{n\to\infty} \sqrt[n]{a_n} = p$:

$$p < 1 \Longrightarrow \sum_{k=1}^{\infty} a_k < \infty,$$

$$p > 1 \Longrightarrow \sum_{k=1}^{\infty} a_k = \infty.$$

2.1.6 Integral Test

$$\int_a^{\infty} f(x)\mathrm{d}x < \infty \Longleftrightarrow \sum_{k=A>a}^{\infty} f(k) < \infty.$$

3 Integral

$$\int f'(g(x))g'(x)\mathrm{d}x = f(g(x))$$

$$\int f(x)g(x)\mathrm{d}x = F(x)g(x) - \int F(x)g'(x)\mathrm{d}x$$

$$\int f(g(x))\mathrm{d}x = \int \frac{1}{g'(x)} f(u)\mathrm{d}u$$

$$g'(x)\mathrm{d}x = \mathrm{d}(g(x))$$

—
- Arc length of $f(x)$ on $[a, b]$:

$$\int_a^b \sqrt{1 + f'(x)^2}\mathrm{d}x$$

- Volume of rotation of $f(x)$ around the x-axis on $[a, b]$:

$$\int_a^b \pi[f(x)]^2\mathrm{d}x$$

- Surface area of the rotated volume:

$$\int_a^b 2\pi f(x)\sqrt{1 + f'(x)^2}\mathrm{d}x$$

—
1. **If the integral contains $a - x^2$:** Use $x = \sqrt{a}\sin u$, where $\mathrm{d}x = \sqrt{a}\cos u\,\mathrm{d}u$, or $x = \sqrt{a}\cos u$, where $\mathrm{d}x = -\sqrt{a}\sin u\,\mathrm{d}u$.
2. **If the integral contains $a + x^2$:** Use $x = \sqrt{a}\tan u$, where $\mathrm{d}x = \frac{\sqrt{a}}{\cos^2 u}\,\mathrm{d}u$.
3. **If the integral contains $x^2 - a$:** Use $x = \sqrt{a}\sec u$, where $\mathrm{d}x = \sqrt{a}\sec u\tan u\,\mathrm{d}u$.
—
Common Integrals Table:
—
This section fully incorporates all the key integrals, substitutions, and formulas from your original text.

4 Integrals and Substitution Rules

$$\int f'(g(x))g'(x)\mathrm{d}x = f(g(x))$$

$$\int f'(x)g(x)\mathrm{d}x = f(x)g(x) - \int f(x)g'(x)\mathrm{d}x$$

Function	**Integral**
$\frac{1}{x^2+1}$	$\arctan x$
$\frac{1}{x^2-1}$	$\frac{1}{2}\ln\left\lvert\frac{x-1}{x+1}\right\rvert$
$\frac{x}{x^2+c}$	$\frac{1}{2}\ln\lvert x^2+c\rvert$
$\frac{1}{\sin x}$	$\ln\left\lvert\tan\frac{x}{2}\right\rvert$
$\frac{1}{\cos x}$	$-\ln\left\lvert\tan\frac{1}{2}\left(x-\frac{\pi}{2}\right)\right\rvert$
$\arcsin x$	$x\arcsin x+\sqrt{1-x^2}$
$\arccos x$	$x\arccos x-\sqrt{1-x^2}$
$\ln x$	$x(\ln x-1)$
$\tan x$	$-\ln\lvert\cos x\rvert$
$\cot x$	$\ln\lvert\sin x\rvert$
$\sin^2 x$	$\frac{x}{2}-\frac{\sin(2x)}{4}$
$\cos^2 x$	$\frac{x}{2}+\frac{\sin(2x)}{4}$
$e^x\sin x$	$\frac{e^x}{2}(\sin x-\cos x)$
$e^x\cos x$	$\frac{e^x}{2}(\sin x+\cos x)$
xe^{kx}	$\frac{e^{kx}}{k}\left(x-\frac{1}{k}\right)$

Table 1: Common integrals for reference.

$$\int f(g(x))\mathrm{d}x=\int\frac{1}{g'(x)}f(u)\mathrm{d}u$$

Arc length of f on $[a,b]$:

$$\int_a^b\sqrt{1+f'(x)^2}\mathrm{d}x$$

Rotation volume of f on $[a,b]$:

$$\int_a^b\pi f(x)^2\mathrm{d}x$$

4.1 Trigonometric Substitutions

If the integral contains $a-x^2$, use $x=\sqrt{a}\sin u$ or $x=\sqrt{a}\cos u$.
If the integral contains $a+x^2$, use $x=\sqrt{a}\tan u$.
If the integral contains x^2-a, use $x=\sqrt{a}\sec u$.

5 Differential Equations

Equation	**Solution**
$y'' + py' + qy = 0$	Let r_1, r_2 be roots of the quadratic: $r^2 + pr + q = 0$. $y = \begin{cases} Ae^{r_1 x} + Be^{r_2 x}, & \text{if } r_1 \neq r_2 \\ e^{rx}(A + Bx), & \text{if } r_1 = r_2 = r \end{cases}$
$y' + f(x)y = g(x)$	$y = \frac{1}{e^{F(x)}} \left(\int e^{F(x)} g(x) \, \mathrm{d}x + C \right),$ where $F(x) = \int f(x) \, \mathrm{d}x$

Table 2: Common forms and solutions for differential equations.

6 Taylor Series

The Taylor series expansion of a function $f(x)$, centered at $x = 0$, is given by:

$$T_N(x) = \sum_{k=0}^{N} \frac{f^{(k)}(0)}{k!} x^k.$$

6.1 Lagrange Error Formula

$$E_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}, \quad c \in [\min(x, a), \max(x, a)]$$

$f(x)$	$T_N(x)$ centered at $x = 0$
e^x	$1 + x + \frac{x^2}{2} + \cdots = \sum_{k=0}^{N} \frac{x^k}{k!}$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots = \sum_{k=1}^{N} \frac{(-1)^{k+1} x^k}{k}$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{k=0}^{N} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{k=0}^{N} (-1)^k \frac{x^{2k}}{(2k)!}$
$\frac{1}{1-x}$	$1 + x + x^2 + \cdots = \sum_{k=0}^{N} x^k$
$\arcsin x$	$x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots = \sum_{k=0}^{N} \frac{(-0.5)^{\underline{k}}}{k!} (-1)^k \frac{x^{2k+1}}{2k+1}$
$\arccos x$	$\frac{\pi}{2} - x - \frac{x^3}{6} - \frac{3x^5}{40} - \cdots = \frac{\pi}{2} - \sum_{k=0}^{N} \frac{(-0.5)^{\underline{k}}}{k!} (-1)^k \frac{x^{2k+1}}{2k+1}$
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} + \cdots = \sum_{k=0}^{N} (-1)^k \frac{x^{2k+1}}{2k+1}$
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \cdots = \sum_{k=0}^{N} \frac{\alpha^{\underline{k}}}{k!} x^k$, where $a^{\underline{b}} = a(a-1)(a-2)\ldots(a-b+1) = \prod_{k=0}^{b-1} (a-k)$

Table 3: Common Taylor series expansions.